Before Worrying About a Gravitational Ghost, Ask Whether It Is Really There

A general-audience guide to building and testing a mathematical universe

Einstein's theory of relativity turns gravity into the geometry of spacetime. Matter and energy curve spacetime; that geometry tells matter, light and clocks how to move.

This picture explains phenomena that are part of everyday science: clocks run at different rates in different gravitational fields, light bends and changes color, gravitational waves cross the universe, black holes form, and the universe expands.

Our project asks whether a different mathematical theory of spacetime—called conformal or pure-Weyl gravity—can support a physically coherent universe. The theory uses equations with four derivatives instead of Einstein's two. That makes it attractive in some approaches to quantum gravity, but it also produces extra solutions whose signs look wrong. These are commonly called gravitational ghosts.

We do not begin by declaring those extra solutions physical or unphysical. We build the mathematical universe in layers and make each layer pass the standard tests physicists use for spacetime, causality, waves, clocks, interactions and quantum particles.

The basic principle is:

> A pattern in an equation is not yet a physical object. It must survive the constraints, carry the appropriate physical pairing or charge, propagate causally, and remain consistent when interactions and quantization are included.

What does it mean to build a universe mathematically?

A candidate theory of gravity needs much more than an elegant equation. Different familiar phenomena have different mathematical acceptance tests.

Spacetime and curvature

First, there must be an exact spacetime geometry that satisfies the field equations. Small disturbances around it must also satisfy a consistent set of linearized equations and constraints.

This is the layer that gives meaning to distances, durations, light cones and curvature. Without it, phrases such as "light ray" or "gravitational wave" have no defined background on which to live.

Cause and effect

A disturbance introduced in one region should affect only places inside its causal future. Mathematically, this is established using retarded and advanced Green operators: one propagates effects forward and the other backward while respecting the spacetime light cones.

For a gauge theory such as gravity, it is not enough to find a Green operator for one convenient field component. The propagation must be compatible with the full system of fields, gauge transformations, equations, constraints and identities.

Light

Classical light requires Maxwell's electromagnetic equations, a causal light cone and nontrivial electromagnetic wave solutions. Quantum particles of light require more: a quantum state space, a positive probability rule and a controlled definition of incoming and outgoing photons.

The existence of a classical electromagnetic wave therefore does not by itself prove the existence of a healthy photon.

Gravitational waves

A genuine gravitational wave must solve the linearized gravitational equations, survive coordinate and scale redundancies, carry a nondegenerate symplectic pairing, and propagate causally. To describe observations such as LIGO signals, the theory must eventually supply boundary conditions, waveforms, energy flux and interactions with detectors.

Clocks, time dilation and redshift

Relativity compares physical clocks, not an abstract coordinate called "time." A clock field must evolve monotonically, have healthy kinetic energy and couple consistently to gravity.

Gravitational redshift requires two such clocks, an emitted light signal and a received light signal. The prediction is the ratio of the measured frequencies. A curved metric and a clock field provide the ingredients, but a redshift theorem requires the complete relational observable connecting emitter, light ray and receiver.

Interactions and stability

Free waves are only the beginning. The nonlinear equations must say how waves scatter and exchange energy. A sector that looks healthy at linear order may fail because interactions generate a forbidden mode, violate a constraint or create an obstruction at the next order.

Quantum particles

Particles require the quantum theory. For gravity this includes a consistent quantum gauge identity, control of anomalies, suitable short-distance states, a physical inner product and, for scattering, a meaningful asymptotic region.

These tests form our universe-building ladder. Passing one rung does not silently grant the rungs above it.

The boundary-free universe we study

The cleanest complete setting models an entire closed universe at once rather than cutting out a region and giving it an artificial outer edge.

At each moment, space is a three-dimensional sphere. It is finite but has no boundary. The closest everyday analogy is the two-dimensional surface of the Earth: it has a finite area, yet a traveler can keep moving without reaching an edge. A three-dimensional sphere is the same closed idea one dimension higher.

Let that whole spherical universe evolve through time. Stacking its successive spatial spheres produces the spacetime mathematicians call the conformal cylinder. The word *cylinder* describes the history of a closed universe; it does not mean that the universe sits inside a pipe or has a cylindrical spatial shape.

This gives an exact model of a complete universe without spatial boundary conditions. The gravitational field separates into a discrete collection of modes, and all fifteen conformal transformations fit into one geometric structure.

Whether the observed universe has this global shape is an empirical question. For this project, the spherical universe is valuable because it exposes the complete constraint system without adding an exterior region where boundary charges can enter.

That last point matters. A transformation can be gauge in one physical sector and carry a measurable charge in another. The words "time is gauge" are not a principle we assume; they are a statement that must be calculated.

Why gauge symmetry changes the ghost question

The same place on Earth can be described using many maps. Coordinates change, but the place does not. Gravity has the same feature: different coordinate descriptions can represent the same spacetime geometry. Conformal gravity has an additional freedom to change the local scale.

A raw list of solutions therefore overcounts physical possibilities.

BRST and BV theory provide a systematic accounting system. They place fields, gauge transformations, equations, constraints, identities and their dual partners into one mathematical complex. The calculation asks:

- Does an object satisfy all gauge constraints?
- Is it merely a gauge transformation of another object?
- Does it carry a charge that prevents us from treating it as gauge?
- Does it have a nonzero physical pairing after reduction?

What survives is called cohomology. In ordinary language, it is the part of the theory that remains after every declared redundancy has been removed.

The bookkeeping variables used by BRST are also called ghosts. They are not the negative-energy particles associated with higher derivatives. One is a tool for tracking gauge symmetry; the other is a possible physical pathology.

The current closed-universe result

Free pure-Weyl gravity on the spherical universe has a complete classical causal BV–BFV description in the selected closed, zero-charge sector.

The full prolonged system contains 386 linked rows of fields, gauge data, equations, identities, curvature variables and auxiliary variables. Retarded and advanced chain homotopies exist on every row with causal support. Their pairing agrees with the covariant current pairing and with the pairing of the reduced state model.

This means that the residual calculation is not detached algebra. It is connected to the actual metric field, its equations and its causal propagation.

The time-translation question has a sector-dependent answer:

- On the unrestricted compact phase space, translation along the universe's time direction carries a nonzero charge. It is a physical symmetry there.
- On the full Taub or moment-map zero sector used by the closed-universe quotient, that charge vanishes and time translation acts as gauge.

When all fifteen residual conformal transformations are imposed as constraints in that zero-charge sector, no individual one-particle conformal-graviton class survives. The calculation does not turn a negative-sign graviton into a positive-sign graviton. The entire one-particle sector is absent from this particular residual cohomology.

The first surviving objects are two composites made from pairs of Weyl curvatures, one for each chirality or handedness. Their combinations correspond to the square of the Weyl curvature and the Pontryagin density. Their normalized pairing is the two-dimensional identity matrix.

These are positive-paired deformation or vertex classes. They are possible directions in the space of actions, interactions or observables. They are not two graviton particles flying through space.

The result is therefore precise but sector-specific:

In the selected boundary-free, zero-charge universe, the physical object produced by the complete free residual reduction is not a one-graviton state space. It is a two-dimensional space of curvature-square deformation classes.

It does not follow that time translation is gauge in a universe with an external boundary, an asymptotic region or a retained Hamiltonian charge.

Clocks and the meaning of time

A useful universe must contain physical reference systems. The clock model uses a Berger version of the spherical universe, in which the spatial three-sphere is smoothly squashed rather than perfectly round. Two standard-sign rotating scalar fields support the geometry and provide a monotone internal phase.

The clock has genuine matter momentum and healthy classical signs. At fixed couplings and linear order, the full constraint equations force the variation of the total time-translation charge to vanish. Thus nontrivial internal clock motion can coexist with total time translation acting as gauge in this declared sector.

This is the mathematical ingredient needed for relational time, but it is not yet a redshift calculation. A certified redshift requires an explicit observable of the form “the received light frequency when the receiver’s clock reads this value,” together with the emitter, the light path and both clock responses.

Light and gravitational waves

The programme contains two related but distinct wave results.

On the pure-Weyl spherical universe, the full free gravitational gauge complex has causal propagation. This establishes a consistent classical response problem for the complete constrained field theory.

On a compact Einstein–Maxwell product universe, the familiar standard electromagnetic and gravitational harmonic modes occur inside the larger Weyl–Maxwell solution space. The complete standard Einstein–Maxwell linear tangent injects into the Weyl–Maxwell theory before the final global quotient, and its induced symplectic pairing is nondegenerate.

This is a real classical light-and-gravitational-wave result. It says that the larger theory contains the conventional linear wave directions rather than discarding ordinary gravitational physics altogether.

It is not yet a particle or detector theorem. The Weyl–Maxwell pairing is not simply identical to the Einstein–Maxwell pairing; its radiative blocks are relatively indefinite. The additional fourth-order wave branch is being classified, and asymptotically flat radiation, Bondi energy and scattering remain separate tests.

There is no contradiction between the presence of these classical wave modes and the absence of one-particle classes in the selected pure-Weyl residual cohomology. They are different backgrounds, matter contents, charge sectors and quotients. The project records those choices explicitly so that results from one setting are not silently promoted to another.

Interactions: does the construction remain a universe?

The clock-coupled Berger calculation includes the complete quadratic interaction operation. After exact transfer it contains 54,236 canonical nonzero coefficients. Its cyclicity identities pass, and the corresponding time-translation Cartan contraction exists through arity two on the complete 54-row classical complex.

In ordinary language, the gauge interpretation of total time translation survives the first nonlinear consistency test in this sector.

That is not full nonlinear stability. Cubic and higher identities, the first physical resonant channels, global evolution and the quantum theory remain independent gates. Other higher-derivative models in the programme exhibit explicit interaction obstructions, so no free positive structure is assumed to survive merely because it works at linear order.

The Einstein–Maxwell sector supplies the complementary warning. Standard linear photon and gravitational directions exist, but explicit compact fixed-charge modes can encounter second-order Taub obstructions. Linear inclusion does not automatically imply nonlinear closure.

The code is a symbolic physics laboratory

The calculations are too large and too sign-sensitive to manage reliably as hand algebra alone. We use code to perform exact symbolic derivations while keeping the mathematical and physical assumptions visible.

The code does not replace the theory with a numerical simulation. It derives and checks algebraic consequences of the equations:

- differential operators are generated from the action and frozen conventions;
- tensor, BRST, BV and homological identities are reduced using exact rational or algebraic arithmetic;
- large transferred interactions are expanded coefficient by coefficient;
- ranks, kernels, pairings, cyclic signs and obstruction classes are computed exactly rather than estimated with floating-point tolerances;
- causal and support claims are separated from purely local or mode-based identities.

Every material claim carries a reproducibility and accountability trail:

1. **Declared scope.** The theory, background, generator, phase space, boundary conditions and lifecycle level are named.
2. **Machine-readable certificate.** Inputs, assumptions, hashes, outputs and unresolved fields are recorded.
3. **Independent verifier.** A separate program replays the decisive identity instead of trusting the generating script.
4. **Failure guards.** Deliberately broken inputs must fail, preventing a verifier from passing vacuously.
5. **Dependency tags.** A local algebra calculation cannot be presented as a Lorentzian causal or quantum theorem.
6. **Clean reproduction audit.** Publication artifacts are rebuilt from a clean archived commit and compared with their recorded hashes.
7. **Negative-result ledger.** Failed ansatzes and scoped no-go theorems remain visible instead of disappearing when a different route succeeds.

This makes the repository closer to a symbolic experimental laboratory than to a conventional collection of informal calculations. A result can be inspected at three levels: the physical statement, the mathematical identity and the exact computational receipt.

The code still does not prove assumptions that were never encoded. Analytic existence, global boundary conditions and quantum interpretation must each receive their own certificates. The system is designed to fail closed when a required layer is missing.

A current map of the universe

The certified construction contains:

- exact curved spacetime backgrounds without spatial boundaries;
- a complete free causal gauge complex for pure-Weyl gravity on the spherical universe;
- an exact distinction between charged time translation and the zero-charge sector where it is gauge;
- two positive-paired residual curvature-square deformation classes;
- a healthy classical matter clock in a fixed-coupling linear Berger sector;
- conventional linear electromagnetic and gravitational wave directions in the compact Einstein–Maxwell inclusion;
- a quadratic clock-coupled interaction operation satisfying the declared cyclic and Cartan identities.

The certified construction does not yet contain:

- an explicit relational gravitational-redshift observable;

- a general theorem on arbitrary globally hyperbolic spacetimes;
- an asymptotically flat Bach phase space with Bondi or ADM charges;
- a complete classification and positive interpretation of the extra fourth-order radiative branch;
- black-hole horizons and their physical boundary conditions;
- full nonlinear gravitational evolution;
- a restored quantum master equation and anomaly-free quantum theory;
- a positive graviton or photon Fock space;
- a scattering matrix, dark-matter prediction or dark-energy cosmology.

Those are not decorations to be asserted at the end. Each is a rung with a recognized mathematical test.

The question the project is designed to answer

The higher-derivative ghost problem is not one yes-or-no calculation. Its answer can change when the background, boundary conditions, charge sector, interactions or quantum theory change.

The central question is:

> Is the apparent extra gravitational branch a physical instability, a gauge direction, a constrained sector, a boundary excitation or a quantum anomaly—and which answer applies in each mathematically complete universe?

The project builds enough of each universe to make that question testable. Success means a coherent ladder from spacetime to causal waves, clocks, interactions and particles. Failure means an exact obstruction with a stated scope and a reproducible certificate. Either result teaches us where a candidate theory of gravity can, or cannot, describe the universe we know.